

Digha Tide Harmonic Predictor

A detailed technical literature on the JavaScript harmonic-analysis program, its mathematics, data, calibration, accuracy, limitations and future development

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Abstract

This document describes the development of a browser-based Digha tide predictor built in JavaScript. The program began as a fixed harmonic simulator in which constituent amplitudes and phases were manually entered. It was then redesigned so that the harmonic coefficients are estimated from the only Digha tide information available for the project: a published seven-to-ten-day-style Tidechart table containing successive high- and low-water times and heights from 19 to 28 August 2026. The available source is not a continuous tide-gauge record; consequently the program is not a professional astronomical tide-prediction engine. Instead, it is best described as a frequency-constrained empirical harmonic extrapolation. The revised program estimates the local coefficients by least squares, compares several candidate constituent sets using leave-one-day-out cross-validation, computes amplitude and phase from the fitted sine/cosine coefficients, and locates high and low water by solving for zeros of the derivative of the fitted tide curve. The selected six-constituent model reproduces the supplied extrema with an in-sample RMSE of about 0.0137 m and a leave-one-day-out RMSE of about 0.0225 m. These numbers describe agreement with the supplied tide-table extrema, not the accuracy of the real ocean on 12 October 2026. The document therefore distinguishes model fit from real-world predictive accuracy and proposes a step-by-step pathway toward a scientifically stronger model using continuous Digha observations, longer records, full astronomical arguments, nodal corrections, more constituents, uncertainty quantification and independent validation.

1. Background: Why Tides Can Be Modelled Harmonically

Tides are a periodic component of coastal water-level variability generated primarily by the gravitational forcing of the Moon and Sun and by the motion of the Earth-Moon-Sun system. The ideal equilibrium tide is not the same as the tide observed at a particular coast. Local bathymetry, coastline geometry, ocean-basin resonance, friction, shallow-water nonlinearities and meteorological forcing alter the amplitude and phase of the observed signal. This is why professional tide prediction is not obtained merely by looking up the Moon's position. NOAA's technical material explicitly notes that observed tides differ appreciably from the idealized equilibrium tide and that local controls prevent prediction from celestial positions alone. Instead, the astronomical forcing is decomposed into periodic tidal constituents and the local response is learned from observations.

A tidal constituent is a periodic component with a known astronomical frequency. Important examples include M2, the principal lunar semidiurnal constituent; S2, the principal solar semidiurnal constituent; N2, a lunar elliptic constituent; K1 and O1, important diurnal constituents; and shallow-water harmonics such as M4. NOAA documentation describes the standard constituent speeds and notes that operational work commonly uses a much larger set of constituents than the six ultimately retained by our small-data model.

The central mathematical insight is Fourier-like. Instead of trying to describe the entire sea-level curve with one wave, the tide is represented as the sum of several waves with different frequencies. Each constituent has a local amplitude and phase. Once these local parameters have been estimated, the model can be evaluated at future times. This is the theoretical basis of harmonic tide prediction.

2. Evolution of Our Program

The first version of the webpage already contained the correct general harmonic form, $\eta(t) = Z_0 + \sum [H_i \cos(w_i t - g_i)]$, and listed standard constituent speeds. However, the original JavaScript manually specified values such as an M2 amplitude of 1.55 m and phase of 95 degrees, an S2 amplitude of 0.62 m and phase of 130 degrees, and similar values for other constituents. The original file itself stated that these amplitudes and phases were calibrated to Digha's general tidal character rather than taken from an official harmonic-constant survey. In other words, that version was a hand-parameterized harmonic simulator rather than a fitted Digha model.

The improved program changes the central philosophy. Instead of manually choosing local amplitudes and phases, it starts with the actual high/low extrema available in the project and estimates the coefficients from them. It also replaces simple sampled maximum/minimum detection with derivative-based extrema detection, adds cross-validation for model selection, uses QR-based least squares rather than explicit matrix inversion, and reports diagnostics. The user interface, interactive curve, date selector, astronomical context and constituent table are retained, but the numerical core is now data-driven.

3. The Actual Data Used

The calibration dataset is the Digha tide table visible in the user's supplied screenshot. It contains 38 successive extrema from 19 August through 28 August 2026. Each record consists of a calendar date, local Indian Standard Time, a water level in metres, and a classification as high or low water. The values include, for example, 19 August at 02:02 with 2.70 m high water, 08:33 with 0.33 m low water, 14:17 with 2.63 m high water and 20:55 with 0.37 m low water. Toward the end of the record, 28 August contains 04:50 at -0.26 m, 10:38 at 3.18 m and 17:10 at -0.21 m. These are the exact source extrema entered into the improved JavaScript model.

This data provenance is crucial. The program is not trained on raw one-minute or five-minute water-level observations. It is calibrated against already-predicted or reported turning points from a tide table. Consequently, the data contain far less information than a continuous tide-gauge record. The word 'trained' should therefore be understood here as 'fitted or calibrated', not machine-learning training. The model is learning the shape of the supplied extrema sequence, not independently discovering the physical ocean response from raw observations.

3.1 Complete Calibration Dataset

Date	Time	Level (m)	Type
2026-08-19	02:02	2.70	High
2026-08-19	08:33	0.33	Low
2026-08-19	14:17	2.63	High
2026-08-19	20:55	0.37	Low
2026-08-20	02:44	2.40	High
2026-08-20	09:10	0.62	Low
2026-08-20	14:58	2.32	High
2026-08-20	21:37	0.62	Low
2026-08-21	03:44	2.11	High
2026-08-21	09:59	0.91	Low

Date	Time	Level (m)	Type
2026-08-21	16:06	2.04	High
2026-08-21	22:35	0.83	Low
2026-08-22	05:40	1.95	High
2026-08-22	11:16	1.11	Low
2026-08-22	18:10	1.91	High
2026-08-23	00:12	0.93	Low
2026-08-23	07:23	2.04	High
2026-08-23	13:32	1.10	Low
2026-08-23	19:39	2.02	High
2026-08-24	02:05	0.77	Low
2026-08-24	08:27	2.26	High
2026-08-24	14:52	0.84	Low
2026-08-24	20:36	2.24	High
2026-08-25	03:04	0.50	Low
2026-08-25	09:11	2.51	High
2026-08-25	15:34	0.55	Low
2026-08-25	21:17	2.48	High
2026-08-26	03:43	0.22	Low
2026-08-26	09:44	2.76	High
2026-08-26	16:07	0.27	Low
2026-08-26	21:50	2.73	High
2026-08-27	04:17	-0.04	Low
2026-08-27	10:13	2.98	High
2026-08-27	16:39	0.01	Low
2026-08-27	22:18	2.96	High
2026-08-28	04:50	-0.26	Low
2026-08-28	10:38	3.18	High
2026-08-28	17:10	-0.21	Low

4. Time Coordinate and Local Time Handling

The browser program works internally in milliseconds since the JavaScript epoch and then converts time to hours relative to a project epoch. The epoch was chosen as 19 August 2026 at 00:00 IST, represented in UTC as 18 August 2026 at 18:30 UTC. This is important because the harmonic phase is meaningful only relative to a clearly defined time origin. The supplied tide times are Indian Standard Time, so the program converts them consistently before constructing the harmonic basis.

For a constituent with angular speed ω expressed in radians per hour, its basis functions are $\cos(\omega t)$ and $\sin(\omega t)$. Using hours rather than days keeps the published constituent speeds in degrees per solar hour convenient to convert. The conversion is $\omega_{\text{rad}} = \text{speed}_{\text{deg_per_hour}} \times \pi/180$. The model therefore has a precise, continuous time coordinate even though the input consists of only discrete high and low events.

5. The Mathematical Model

The original cosine-with-phase form is convenient for presentation:

$$\eta(t) = Z_0 + \sum_i H_i \cos(\omega_i t - g_i)$$

For estimation, however, it is more useful to expand each phase-shifted cosine. Using $\cos(A - B) = \cos A \cos B + \sin A \sin B$, each constituent can be written as a cosine coefficient plus a sine coefficient. The fitted form is therefore:

$$\eta(t) = Z_0 + \sum_i [a_i \cos(\omega_i t) + b_i \sin(\omega_i t)]$$

This transformation is extremely important because Z_0 , a_i and b_i enter the model linearly. Once the frequencies are fixed, the unknown parameters can be estimated using linear least squares. The nonlinear-looking tidal problem is therefore converted into a standard linear regression problem with physically motivated periodic regressors.

6. Design Matrix and Least Squares

Suppose there are N observed extrema and k tidal constituents. For each observation at time t_j , the program constructs a row containing one intercept column followed by cosine and sine columns for every constituent. For a two-constituent example the row is:

$$[1, \cos(\omega_1 t), \sin(\omega_1 t), \cos(\omega_2 t), \sin(\omega_2 t)]$$

Stacking all rows produces the design matrix X . The observed heights form the vector y and the unknown coefficients form β . The model is $y \approx X \beta$. Ordinary least squares chooses β to minimise the sum of squared residuals:

$$SSE = \sum_j (y_j - \hat{y}_{tj})^2$$

The familiar closed-form expression $\hat{\beta} = (X'X)^{-1} X'y$ is mathematically valid when the matrix is well conditioned, but explicitly forming the inverse is not the best numerical practice. The improved program therefore solves the least-squares problem with a Householder QR procedure. QR factorisation is a standard stable approach because it avoids directly forming $X'X$ and is less vulnerable to the loss of numerical precision caused by squaring a condition number.

7. Recovering Harmonic Amplitude and Phase

After least squares estimates a_i and b_i , the familiar amplitude-phase form is recovered. The constituent amplitude is:

$$H_i = \sqrt{a_i^2 + b_i^2}$$

The phase convention used by the program is obtained with the two-argument arctangent:

$$g_i = \text{atan2}(-b_i, a_i)$$

The phase is then converted to degrees and wrapped to the interval 0–360 degrees. This is preferable to using a one-argument arctangent because atan2 preserves the quadrant and therefore gives the correct phase orientation.

8. Constituent Frequencies Used

Constituent	Physical description	Speed (°/solar hour)	Fitted amplitude (m)	Fitted phase (°)
M2	Principal lunar semidiurnal	28.9841042	1.2731	266.66
S2	Principal solar semidiurnal	30.0000000	0.6478	354.44
N2	Larger lunar elliptic semidiurnal	28.4397295	0.1799	145.05
K1	Lunisolar diurnal	15.0410686	0.0815	163.85
O1	Principal lunar diurnal	13.9430356	0.0416	318.48
M4	Shallow-water overtide of M2	57.9682084	0.0285	33.82

The speeds are not fitted from the ten-day extrema record. They are fixed constituent frequencies. This distinction matters: the data estimate the local response coefficients, while the astronomical frequencies are imposed from tidal theory. NOAA reference material gives, for example, $M2 = 28.984104^\circ/\text{h}$, $S2 = 30^\circ/\text{h}$, $N2 = 28.43973^\circ/\text{h}$, $K1 = 15.041069^\circ/\text{h}$ and $O1 = 13.943035^\circ/\text{h}$. The fitted amplitudes and phases in this table are our model's coefficients, not official Digha harmonic constants.

9. Why Only Six Constituents?

The program considers five candidate models: $M2+S2$; $M2+S2+N2$; $M2+S2+N2+M4$; $M2+S2+N2+K1+O1$; and $M2+S2+N2+K1+O1+M4$. The final model was selected by leave-one-day-out cross-validation. This is deliberately conservative. A model with more parameters can almost always reduce in-sample error, but with only 38 extrema, adding many closely spaced constituents would create an unstable and poorly identified model. Professional tide analysis commonly uses substantially longer records and many more constituents; the small model here is a compromise between physical richness and identifiability.

The six selected constituents also have a sensible physical interpretation. $M2$ and $S2$ supply the dominant semidiurnal cycle; $N2$ modifies the lunar semidiurnal pattern; $K1$ and $O1$ represent major diurnal contributions; and $M4$ provides a shallow-water overtide that can help represent the non-sinusoidal shape of a coastal tide. The program deliberately does not claim that these six are the complete Digha spectrum.

10. Model Selection and Cross-Validation

A key improvement over the original hand-tuned program is that model complexity is selected using leave-one-day-out validation. There are ten calendar dates in the calibration record. For each candidate model, the program removes one complete day, fits the model to the remaining nine days, predicts all extrema on the held-out day, and computes the errors. This is repeated for every day. The final cross-validation RMSE is the square root of the average squared held-out error.

Model	In-sample RMSE (m)	R ²	Leave-one-day-out RMSE (m)
$M2 + S2 + N2 + K1 + O1 + M4$	0.0137	0.999827	0.0225
$M2 + S2 + N2 + K1 + O1$	0.0176	0.999715	0.0232
$M2 + S2 + N2$	0.0746	0.994892	0.0809
$M2 + S2 + N2 + M4$	0.0704	0.995451	0.0815
$M2 + S2$	0.0792	0.994232	0.0869

The selected model is $M2 + S2 + N2 + K1 + O1 + M4$. Its in-sample RMSE is 0.0137 m, R^2 is 0.999827, and leave-one-day-out RMSE is 0.0225 m. The cross-validation criterion is the reason the final model is not simply the largest candidate model available.

11. Finding High and Low Tide Times

Once the fitted function $\eta(t)$ exists, a high or low tide is a local extremum. Mathematically, an extremum occurs when the first derivative with respect to time is zero. Differentiating the fitted model gives:

$$d\eta/dt = \sum_i [-a_i w_i \sin(w_i t) + b_i w_i \cos(w_i t)]$$

The browser program scans the day at one-minute intervals to identify sign changes in the derivative. Each sign change brackets a root. The root is then refined by bisection for much higher precision. The second derivative is estimated numerically around the root. A negative second derivative identifies a

local maximum (high water); a positive second derivative identifies a local minimum (low water). This is mathematically stronger than simply selecting the largest sampled height from a one-minute grid.

12. 12 October 2026: Model Output

Predicted event	Local IST	Height (m)
Low	12 Oct 2026, 05:38:11 AM	-0.327
High	12 Oct 2026, 11:52:40 AM	3.351
Low	12 Oct 2026, 05:56:14 PM	-0.409
High	13 Oct 2026, 12:11:02 AM	3.296

The model therefore produces approximately 05:38 AM low water at -0.327 m, 11:53 AM high water at 3.351 m, 05:56 PM low water at -0.409 m, and the following high water at approximately 12:11 AM on 13 October at 3.296 m. The last event is outside the strict 00:00–24:00 window of 12 October but is shown because it completes the tidal cycle across midnight.

13. What Does 'Accuracy' Mean Here?

It is tempting to report the 0.0137 m RMSE and conclude that the program predicts Digha tides to within roughly 1.4 cm. That conclusion would be scientifically incorrect. The RMSE measures how closely the fitted mathematical curve reproduces the 38 input extrema. Those extrema themselves came from a tide table rather than from independent raw observations. The model can therefore fit the published sequence very well while still having an unknown error when extrapolated to a future date.

The leave-one-day-out RMSE is more informative. Its value of approximately 0.0225 m means that when one calendar day is removed, the model trained on the remaining nine days predicts the omitted extrema with an average root-mean-square error of about 2.25 cm. This is a genuine out-of-sample test within the same short dataset, so it is evidence that the harmonic structure generalises across nearby dates. However, it is still not an independent real-world accuracy assessment because all ten days come from the same published tide-table source and the validation period is very close to the calibration period.

A proper accuracy statement would require an independent continuous observed water-level record. The fitted model would be trained on one period and evaluated against observations from a completely separate period, ideally several months or a year. Metrics should include RMSE, mean absolute error, maximum absolute error, high-water timing error, low-water timing error, and perhaps separate errors for spring and neap conditions.

14. Why the 50-Minute Rule Was Removed

Earlier in the project we used the intuitive rule that successive tides shift by roughly 50 minutes per day because a lunar day is about 24 hours 50 minutes. This is useful for understanding the phenomenon, but it is not a professional prediction algorithm. The actual tide signal is the superposition of multiple constituents with different frequencies and phases. Their interference changes the exact timing and range from day to day. The revised program therefore never takes an August tide time and simply adds 50 minutes multiplied by the number of days. Instead, it evaluates the fitted harmonic function directly at the future time.

15. Why Moon Phase and Sunrise Are Not Direct Inputs

The webpage displays Moon phase, illumination, sunrise and sunset because these are useful astronomical context. They are not directly inserted into the fitted tide equation. This is intentional. Simply adding a 'Moon phase factor' to the water level would not reproduce the professional harmonic method. The astronomical information belongs in the calculation of constituent arguments and nodal corrections in a full tide-prediction system. In our simplified model, the astronomical frequencies are fixed and the local coefficients are estimated from the available tide data.

16. Major Scientific Limitations

- The calibration record contains only 38 extrema over 19–28 August, not continuous water-level observations.
- The input tide table may itself be a model prediction; fitting it does not independently validate the physics.
- The record is too short to reliably estimate a large set of closely spaced tidal constituents.
- The model does not implement full astronomical arguments, nodal amplitude corrections or equilibrium arguments.
- The model does not explicitly represent meteorological residuals such as wind setup, atmospheric pressure, rainfall or freshwater discharge.
- The height datum is inherited from the supplied source and should not be interpreted as a rigorously established hydrographic datum.
- The model is local to the supplied data and should not be transferred to another station.
- The excellent in-sample fit is partly expected because the model is fitted to the same extrema it is designed to reproduce.

17. How the Program Can Be Improved with Real-World Data

The most important future improvement is to replace the extrema-only calibration set with a continuous Digha tide-gauge record. The ideal input would be timestamped water level at one-minute or five-minute intervals for at least 29–30 days as a first serious analysis, with several months or a full year being much better. A continuous record contains the entire waveform between high and low water and therefore provides thousands or hundreds of thousands of observations rather than only a few turning points.

Step 1 — Acquire genuine observations. Obtain a Digha tide-gauge time series from an authoritative oceanographic or hydrographic source. Preserve the original timestamps, units, datum, station identifier and quality flags.

Step 2 — Standardise time. Convert every timestamp to UTC internally, retain the original IST representation for display, and check daylight-saving issues if data from other regions are ever added.

Step 3 — Quality control. Detect missing records, duplicate timestamps, impossible jumps, sensor spikes and long gaps. Do not silently interpolate long gaps. Short gaps can be handled with carefully documented interpolation or by excluding affected observations.

Step 4 — Establish the vertical datum. Determine exactly what the measured height is referenced to. Convert to a consistent datum before comparing the model with published tide heights.

Step 5 — Build a larger harmonic basis. Start with M2, S2, N2, K1 and O1. Then add P1, Q1, K2 and appropriate shallow-water constituents only when the record length supports them.

Step 6 — Fit the coefficients. Use least squares on the continuous observations. Prefer QR or singular-value decomposition over explicit normal-equation inversion.

Step 7 — Add full astronomical arguments. Instead of using only fixed angular speeds, calculate the equilibrium arguments of the astronomical constituents for the requested date and apply the standard nodal corrections.

Step 8 — Validate independently. Hold out a later month or year that was never used for fitting. Compare predicted water levels and predicted high/low events with observations.

Step 9 — Quantify uncertainty. Estimate confidence intervals or covariance for fitted coefficients and propagate parameter uncertainty to predicted heights and event times.

Step 10 — Compare against official predictions. For a known date, compare the program with an authoritative tide table. Differences should be diagnosed rather than simply hidden by recalibration.

18. A Research-Grade Version of the Project

With sufficient observations, the project could become a real scientific tide-analysis package. The JavaScript frontend could remain the visualisation layer while a C, Python or WebAssembly numerical core performs the harmonic analysis. A robust architecture would separate data ingestion, quality control, astronomical calculations, harmonic estimation, prediction, event detection, validation and visualisation.

The estimation engine could support weighted least squares, where observations with known larger uncertainty receive lower weight. Singular-value decomposition could be added to diagnose nearly collinear constituents. Automatic constituent selection could be based on cross-validation, information criteria and physical significance. The program could also estimate confidence intervals for each amplitude and phase and propagate them to uncertainty bands around the future tide curve.

A particularly valuable extension would be a hindcast/forecast framework. For example, fit the model to January–September data and predict October; then compare every predicted high and low with the October observations. Repeat this through the year. Such rolling-origin validation would reveal whether the model remains stable across spring-neap cycles, seasonal changes and unusual meteorological periods.

19. Possible Future Research Questions

- How much does the prediction improve when 10 days, 30 days, 90 days and 365 days of continuous data are used?
- How many constituents can be reliably resolved as the record length increases?
- How much of Digha's tidal range is explained by M2 and S2 alone?
- How strongly do K1 and O1 explain the observed diurnal inequality?
- Does adding M4 materially improve the shape and timing of Digha's shallow-water tide?
- How stable are the estimated Digha harmonic constants from one season to another?
- How much of the residual error is meteorological rather than tidal?
- Can wind and atmospheric-pressure data explain short-term residuals after harmonic prediction?
- How closely does a locally fitted model agree with an authoritative INCOIS or hydrographic prediction?

- Can the model be generalised to other Bengal coast stations and used to study spatial variation in tidal response?

20. Reproducibility and Software Design

The improved program is intentionally self-contained in JavaScript so that it can be opened directly in a modern browser. Its numerical core contains the calibration dataset, fixed constituent frequencies, QR least-squares solver, model-selection loop, amplitude/phase conversion, derivative calculation, bisection-based root refinement and graphical rendering. This makes the calculation transparent and auditable: a reader can inspect the source rather than relying on an opaque external API.

The program also separates model output from astronomical context. The date selector changes the prediction window; the harmonic function is evaluated continuously; the event detector identifies local maxima and minima; and the diagnostic panel reports how the chosen model was selected. This design makes the webpage suitable as an educational research prototype and as a foundation for a future full harmonic analysis engine.

21. Overall Assessment

The current program should be understood as a strong educational and computational prototype rather than a certified navigation-grade tide predictor. Within the severe data limitation of having only a short sequence of published high/low extrema, the model does something substantively better than the original hand-tuned simulator: it estimates its local harmonic coefficients from the available Digha data, selects model complexity by out-of-sample validation, and finds event times from the mathematical derivative. Its very low validation error shows that a compact harmonic representation captures most of the short-term pattern in the supplied tide table.

The most important scientific conclusion is therefore not that the program is 'accurate to 2.2 cm'. The defensible conclusion is that the six-constituent empirical harmonic model provides a close internal reconstruction and short-range extrapolation of the supplied Digha tide-table pattern. Its true future accuracy remains unknown until it is compared with independent observations. The next major scientific step is consequently data acquisition and validation, not simply adding more equations.

22. References and Technical Reading

NOAA Tides & Currents — Tidal Analysis and Prediction. Technical reference describing harmonic analysis, constituent prediction and tidal prediction methodology.

https://tidesandcurrents.noaa.gov/publications/Tidal_Analysis_and_Predictions.pdf

NOAA Tides & Currents — Manual of Harmonic Analysis and Prediction of Tides. Detailed treatment of the astronomical tide-producing forces, harmonic analysis and prediction.

<https://tidesandcurrents.noaa.gov/publications/SpecialPubNo98.pdf>

NOAA Tides & Currents — Understanding Tides. Educational/technical treatment of constituent speeds, physical origins and the multi-constituent nature of tides.

https://tidesandcurrents.noaa.gov/publications/Understanding_Tides_by_Steacy_finalFINAL11_30.pdf

NOAA Tides & Currents — Harmonic Constituents. Examples of station-specific harmonic constants and the standard constituent set.

<https://www.tidesandcurrents.noaa.gov/harcon.html>

NOAA Tides & Currents — Frequently Asked Questions. Discussion of tide-prediction accuracy and the distinction between astronomical tide and meteorological water-level effects.

<https://www.tidesandcurrents.noaa.gov/faq.html>

Source dataset for this project. User-supplied Digha Tidechart screenshot containing the 19–28 August 2026 high/low tide extrema used for calibration.
Provided in the project conversation; not an independent raw tide-gauge dataset.

Final note: This document intentionally distinguishes observed/source data, fitted quantities, model assumptions and external scientific reference material. The program's fitted constants are project-specific coefficients and should not be represented as official Digha harmonic constants unless independently derived from authoritative tide-gauge observations.